



Intersonic shear crack propagation using peridynamic theory

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Abstract Dynamic crack propagation of mode-II cracks is simulated using bond-based Peridynamic Theory (PD) implemented in finite element analysis software ABAQUS. The specimen is a bonded homogeneous Homalite plate with a pre-notch that is subjected to impact shear loading simulating the experiments of Rosakis et al. (1999). The PD bonds at the bonding interface are utilized with a scalar critical stretch value that corresponds to mode-II fracture toughness of the interface. The crack initiation and propagation are naturally captured in the bond-based PD simulations by modifying the original prototype microelastic brittle law formulation introduced by Silling and Askari (2005). Impact loading is introduced at the specimen as a pulse speed field boundary condition. Using bond-based PD, sub-Rayleigh and intersonic regimes

of crack growth are obtained as a function of fracture toughness (G_{II}) and impact speed (V_i) values. The intersonic crack growth is discerned from the sub-Rayleigh crack growth by the existence of shear Mach waves in the particle velocity magnitude contours. For critical values of G_{II} and V_i , a crack growing at a speed just below the Rayleigh wave speed is observed to transition to an intersonic speed with a Burridge-Andrews mechanism. The sustained intersonic crack tip speed is found to be between $1.57c_S$ (c_S is the shear wave speed) and c_D (c_D is the dilatational wave speed). For a reduced impact pulse duration, an intersonic crack is found to approach the theoretical value of $\sqrt{2}c_S$, which, however is not maintained. The results are in qualitative agreement with the experiments of Rosakis et al. (1999) and previous simulations in the literature.

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1 Introduction

The limit of attainable crack speeds for dynamically propagating cracks in homogenous solids can be derived from the elastic solutions of dynamic energy release rates for different velocity regimes and loading conditions. As explained by Freund (1998) and Broberg (1999), the dynamic energy release rate monotonically reduces to zero as the crack tip speed increases from zero to the Rayleigh wave speed, c_R , for mode-

I cracks propagating in a straight path. The analytical solutions of the dynamic energy release rate yields negative results for crack tip speeds above c_R which indicate that mode-I crack propagation at speeds above c_R is not possible since crack propagation is an energy consuming process (Broberg 1989). Experimental studies of dynamic mode-I cracks show that crack tip speeds of mode-I cracks remain below half of the Rayleigh wave speed, c_R , in a homogeneous isotropic solid (Ravi-Chandar and Knauss 1984). In the experiments of Xu and Needleman (1994) and Sharon and Fineberg (1996), crack path instabilities and crack branching is observed at crack tip speeds below c_R , which makes the theoretical speed limit c_R inaccessible in homogenous isotropic solids. Lee and Knauss (1989) and Washabaugh and Knauss (1994) demonstrated that mode-I crack tip speeds asymptotically reach the theoretical limit c_R if the crack path is restrained to weak plane along two identical solids.

The admissible speed regimes of mode-II cracks propagating on a prescribed straight path can be analytically determined based on energetic considerations. For mode-II loading conditions, the value of the dynamic energy release rate is finite and positive for crack speeds either below the Rayleigh wave speed, c_R , (sub-Rayleigh) or between the shear wave speed, c_S , and dilatational wave speed, c_D , (intersonic) (Broberg 1989). Freund (1979) showed that $\sqrt{2}c_S$ is the only stable intersonic speed for dynamic mode-II cracks in homogenous linear elastic materials based on the asymptotic solution of singular stress fields around an intersonic mode-II crack. Andrews (1976) investigated the propagation of intersonic mode-II cracks on a pre-defined straight path using a slip weakening cohesive relation and concluded that sustained shear crack tip speeds can be obtained below c_R or slightly above $\sqrt{2}c_S$ depending on the cohesive strength. Burridge et al. (1979) investigated the stability of mode-II cracks numerically and concluded that stable mode-II crack propagation is possible for crack speeds either below c_R or between $\sqrt{2}c_S$ and c_D . Abraham and Gao (2000) performed molecular dynamics simulations of mode-II crack propagation along a weak plane and observed that microcracks nucleate ahead of the shear crack front as the crack tip speed reaches c_R . They found that, the crack tip speed rapidly jumps to intersonic regime and sustains around $\sqrt{2}c_S$ as the microcracks and the main crack tip coalesce. Inter-sonic crack tip speed of $\sqrt{2}c_S$ was attained when the applied pulse was completely

released as soon as the main crack tip and microcracks merged (Abraham and Gao 2000). Otherwise, the sustained crack tip speed stayed at a higher crack tip speed value between $\sqrt{2}c_S$ and c_D .

Pure mode-II crack propagation on a straight path can be experimentally investigated by generating pre-defined weak planes where the fracture toughness is significantly lower than the bulk materials. Thus, experimental studies of inter-sonic cracks were conducted using bonded bi-material specimens where the crack propagates along the interface (Liu et al. 1993; Lambros and Rosakis 1995; Rosakis et al. 1998). Rosakis et al. (1999) conducted the first experimental study of inter-sonic shear crack growth on isotropic materials subjected to in-plane shear impact loading. They suppressed the kinking of the crack path by gluing two pieces of Homalite plates that generate a weak interface plane. The experimental study of Rosakis et al. (1999) showed that mode-II crack speed can exceed the Rayleigh wave speed and crack propagation is stable at a speed just above $\sqrt{2}c_S$ in the inter-sonic regime. The experiments of Rosakis et al. (1999) was simulated by Needleman (1999), Daphalapurkar et al. (2007) and Samudrala et al. (2002) using cohesive zone formulations. The lowest inter-sonic stable crack tip speed was found to be higher than $\sqrt{2}c_S$ in Needleman (1999); Daphalapurkar et al. (2007); Samudrala et al. (2002). Coker and Rosakis (2001) conducted asymmetric impact experiments using unidirectional graphite epoxy composite specimen and observed shear dominated inter-sonic crack growth along the fibers which create a natural weak interlayer. Later on, Yu et al. (2002) simulated the experiments conducted by Coker and Rosakis (2001) using an anisotropic cohesive law.

Classical Continuum Mechanics (CCM) based tools such as Finite Element Method (FEM) are commonly used in the literature for numerical simulation of dynamic fracture problems (Nishioka 1997). However, there are still major challenges to predict dynamic crack behaviour in the framework of CCM. Equations of motion in CCM consist of spatial derivatives that are not defined at discontinuities by definition. Peridynamic theory (PD) originally developed by Silling (2000) is a robust theory to deal with discontinuities of the structures. It is a non-local theory that employs integro-differential formulation. Thus, it does not require a special treatment at the crack tip and remeshing is not necessary. PD considers non-local interactions between grid points within a region known as horizon H as seen in Fig. 1.

As the radius of horizon, H , goes to zero, PD converges to the classical elasticity theory (Silling and Lehoucq 2008). PD allows both propagation of pre-existing cracks and nucleation of new cracks on completely undamaged regions (Silling et al. 2010). The first formulation of PD assumes pairwise interactions between material points and is named “bond-based” PD (Silling 2000). This assumption leads to a fixed Poisson’s ratio of 1/3 in two-dimensional problems (and 1/4 in three-dimensional problems) (Gerstle et al. 2005). Later, this constraint was eliminated by “state-based” PD that also accounts for indirect interactions between material points (Silling et al. 2007).

PD was successfully applied to a wide range of crack propagation problems in the literature. Ha and Bobaru (2010) analyzed dynamic crack growth of glass specimens under mode-I loading using bond-based PD. They calculated crack tip speeds around half of the Rayleigh wave speed, c_R , and successfully validated their PD results. Agwai et al. (2011) analyzed three different dynamic crack propagation problems using PD and compared the PD estimations of crack paths and speeds with experimental data, Extended Finite Element Method (X-FEM) and Cohesive Zone Modeling (CZM) solutions from the literature. They found that PD estimations capture the experimental crack patterns including microbranches that could not be predicted with X-FEM and CZM methods. Characteristic features of dynamic crack propagation such as crack branching and crack path instabilities were captured with PD (Kilic and Madenci 2009; Ha and Bobaru 2011; Bobaru and Zhang 2015). Fracture in sophisticated patterns due to impact damage in a layered glass plate was analyzed using bond-based PD (Hu et al. 2013). The PD predictions of complex fracture patterns in a multilayered glass subjected to impact in Hu et al. (2013) overlaps with the experimental fracture patterns. The experimentally measured crack tip speed in Polymethyl methacrylate (PMMA) specimen was successfully reproduced with bond-based PD simulations in Mehrmashhadi et al. (2019) where a bilinear PD material model that accounts for the softening behaviour of PMMA is used. Ghajari et al. (2014) proposed an anisotropic material model within the framework of bond-based PD that is suitable for dynamic crack propagation analysis in heterogeneous brittle micro structures. Cheng et al. (2018) derived a bond-based PD model for functionally graded materials (FGM) and investigated the effect of material gradient on crack

branching in FGM. Zhang et al. (2018) performed bond-based PD analyses of edge impact on a brittle ceramic polycrystalline sample and observed inter-sonic propagation of the main damage front and sub-Rayleigh propagation of cracks emanating from the main failure front. Zhou et al. (2018) validated PD simulations of sub-Rayleigh dynamic crack growth problems with experimental results from the literature. They also proposed a modified micromodulus function as a function of the bond stretch and rotation in order to overcome the fixed Poisson’s ratio of bond-based PD.

PD theory can be implemented in various numerical tools in a computationally efficient manner. Macek and Silling (2007) proposed a procedure to implement peridynamic bonds as truss elements to be used in FEA tools. Delamination behaviour of bi-material strip subjected to thermal strains was investigated by Beckmann et al. (2013) using PD Theory implemented in ABAQUS. A bond-based PD element that uses common nodes with conventional FEA elements was developed in Huang et al. (2019). Timoshenko beam and Mindlin plate idealizations of bond-based PD were presented in Diyaroglu et al. (2015); Yolum and Güler (2020) and was implemented in ANSYS by Yang et al. (2019).

In this study, dynamic propagation of a mode-II crack in a Homalite specimen subjected to impact shear loading is investigated using bond-based PD. This paper is organized as follows: a brief review of the PD theory including connections between the PD bond parameters and material properties are presented in Sect. 2. The problem set-up for the asymmetric shear impact problem and material properties of Homalite are explained in Sect. 3. In Sect. 4, a PD mesh convergence study is initially conducted for inter-sonic crack propagation under impact loading. Then a detailed study in terms of particle velocity contours and crack tip speed history is presented at different impact speeds and fracture toughnesses, highlighting sub-Rayleigh, inter-sonic and transition crack propagation behaviour. In Sect. 5, a parametric study of the effect of impact speed and fracture toughness on the crack tip speed is discussed. Additionally, the effect of the pulse width on the inter-sonic crack tip speed and comparison with results from the literature is presented. The conclusion of the study is presented in Sect. 6.

2 Peridynamic theory

Classical Continuum Mechanics (CCM) is based on the assumption that a body is composed of a continuous material that can be divided into infinite elements with the same physical properties of the bulk material. This fundamental assumption can also be traced to the governing equations of CCM since they are partial differential equations that contain spatial derivatives. The governing equation of motion in CCM is the equilibrium equation which can be written in the following form

$$-\rho(\mathbf{x})\ddot{\mathbf{u}}(\mathbf{x}, t) + \nabla \cdot \boldsymbol{\sigma}(\mathbf{x}, t) + \mathbf{b}(\mathbf{x}, t) = 0, \quad (1)$$

where $\rho(\mathbf{x})$, $\ddot{\mathbf{u}}$ and $\mathbf{b}(\mathbf{x}, t)$ denote the mass density, the spatial acceleration and the prescribed body force density vector of the material point, $\mathbf{x}$, respectively. The solution of Eq. (1) exists only for continuous bodies in space. Thus, the original CCM formulation is not sufficient whenever a defect such as a crack or a void exists in a body. Therefore, analysis of dynamic crack propagation is a major challenge in the framework of CCM based tools such as FEM.

To overcome the shortcomings of CCM in failure prediction, Silling (2000) introduced a non-local theory called Peridynamic Theory (PD) that does not require spatial derivatives. The PD equation of motion can be written as

$$\rho(\mathbf{x})\ddot{\mathbf{u}}(\mathbf{x}, t) = \int_H \mathbf{f}(\mathbf{u}' - \mathbf{u}, \mathbf{x}' - \mathbf{x})dH + \mathbf{b}(\mathbf{x}, t), \quad (2)$$

where $\mathbf{u}$ denotes the displacement vector and $\mathbf{f}$ is the peridynamic force density vector that arise from the interaction of material points located at $\mathbf{x}$ and $\mathbf{x}'$ as illustrated in Fig. 1. These interactions are assumed to be existing between the material points as bonds in the PD literature and they occur within the horizon, H , that has a finite radius of δ as illustrated in Fig. 1. PD theory replaces the spatial derivatives of the stress tensor $\boldsymbol{\sigma}$ in Eq. (1) with the integral of the PD force densities as given in Eq. (2). The integro-differential formulation of PD is an exceptional feature for crack propagation simulations as the PD equation of motion remains valid at discontinues such as cracks or voids.

In the bond-based PD Theory, the bond force vectors $\mathbf{f}$ and $\mathbf{f}'$ at the material points $\mathbf{x}$ and $\mathbf{x}'$ arising from their deformation have equal magnitude in opposite directions in order to assure the conservation of linear momentum as shown in Fig. 1. In addition, the

vectors $\mathbf{f}$ and $\mathbf{f}'$ are parallel to their current relative position vector $(\boldsymbol{\xi} + \boldsymbol{\eta})$ in order to satisfy the conservation of angular momentum. It is also well known that the conservation of linear and angular momentums leads to a constraint on the Poisson's ratio, ν , of the material to be $\nu = 1/3$ for plane stress and $\nu = 1/4$ for plane strain conditions in the bond-based PD theory (Gerstle et al. 2005). There are plenty of engineering materials that have a Poisson's ratio close to $1/3$. Thus, bond-based PD theory can be applied to a variety of dynamic fracture problems despite the constraint on the Poisson's ratio.

In PD theory, the relative position and the relative displacement of the material points $\mathbf{x}$ and $\mathbf{x}'$ in reference configuration is denoted with $\boldsymbol{\xi}$ and $\boldsymbol{\eta}$, respectively, as illustrated in Fig. 1. They can be expressed as

$$\boldsymbol{\xi} = \mathbf{x}' - \mathbf{x}, \quad (3a)$$

$$\boldsymbol{\eta} = \mathbf{u}' - \mathbf{u}. \quad (3b)$$

The energy stored in a single PD bond is expressed in terms of the micropotential function, $\omega(\boldsymbol{\eta}, \boldsymbol{\xi})$, whose unit is the energy per unit volume squared. For microelastic materials, the relation between the micropotential, ω , and the force density, $\mathbf{f}$, is given by Silling (2000) as

$$\mathbf{f}(\boldsymbol{\eta}, \boldsymbol{\xi}) = \frac{\partial}{\partial \boldsymbol{\eta}} \omega(\boldsymbol{\eta}, \boldsymbol{\xi}). \quad (4)$$

For linear elastic materials, micropotential function, ω , in Eq. (4) can be defined as

$$\omega(\boldsymbol{\eta}, \boldsymbol{\xi}) = \frac{cs^2(\boldsymbol{\eta}, \boldsymbol{\xi})|\boldsymbol{\xi}|}{2}, \quad (5)$$

where c denotes the PD bond constant, s , is the stretch between material points $\mathbf{x}$ and $\mathbf{x}'$ and can be calculated as

$$s = \frac{|\boldsymbol{\xi} + \boldsymbol{\eta}| - |\boldsymbol{\xi}|}{|\boldsymbol{\xi}|}. \quad (6)$$

Substituting Eq. (5) in Eq. (4), the peridynamic force density vector, $\mathbf{f}$, can be calculated as

$$\mathbf{f}(\boldsymbol{\eta}, \boldsymbol{\xi}) = f \frac{\boldsymbol{\xi} + \boldsymbol{\eta}}{|\boldsymbol{\xi} + \boldsymbol{\eta}|}, \quad (7)$$

where f is the scalar bond force for linear elastic materials, given by,

$$f = cs. \quad (8)$$

2.1 Constitutive modelling

In order to characterize the elastic behaviour of an isotropic material in bond-based theory, the bond stiffness, c , has to be determined. For microelastic materials, the stiffness of the bonds are assumed to be constant (Silling and Askari 2005). The behaviour of the interactions in the bonds can be thought as springs between the material points (Silling and Askari 2005). The spring constant of a PD bond can be derived in terms of elastic properties of the material (E and ν) by equating the PD strain energy density with the strain energy density obtained from the theory of elasticity under the same loading conditions as in Silling and Askari (2005); Gerstle et al. (2005). In the present study, the two dimensional constitutive model for linear elastic materials under plane stress condition as in Gerstle et al. (2005) is utilized. The PD bond constant, c , for 2D plane stress condition can be defined as

$$c = \frac{12E}{(\pi t \delta^3)(1 + \nu)}, \quad (9)$$

where E and t denote the elastic modulus and the plate thickness, respectively and the Poisson's ratio is $\nu = 1/3$.

In order to define failure, a history dependent damage function, $\mu(t, \xi)$, can be introduced by redefining the force density function, f given in Eq. (8) as

$$f = c s \mu(t, \xi). \quad (10)$$

Prototype microelastic brittle (PMB) damage model derived by Silling and Askari (2005) allows the bonds to break if they are stretched beyond a predefined limit. However, this conventional model does not consider the damage behavior in the compressive states. In mode-II loading conditions, there are both tensile and compressive loaded bonds along the crack path due to anti-symmetric deformation of fracture surfaces (Ren et al. 2016; Zhang and Qiao 2019). In order to capture mode-II crack propagation in PD simulations correctly, both tensile and compressive loaded bonds ahead of the crack tip should have potential to fail. Therefore, we modify the PMB formulation in Silling and Askari (2005) by introducing a negative critical stretch, $-s_c$, as

$$\mu(t, \xi) = \begin{cases} 0, & s(t, \xi) \leq -s_c \\ 1, & -s_c < s(t, \xi) < s_c \\ 0, & s(t, \xi) \geq s_c \end{cases} \quad (11)$$

If the bond stretch is greater than the critical stretch in tension ($s > s_c$), or less than the critical stretch in

compression ($s < -s_c$), the bond between two material points breaks irreversibly according to the modified PMB rule given in Eq. (11) (see Fig. 2). In the literature, similar PD damage laws accounting for bond failure in the compressive region are employed to capture compressive and shear failures in Gerstle et al. (2007); Du et al. (2020). Damage accumulation at a material point, $\mathbf{x}$, can be calculated as (Silling and Askari 2005):

$$\varphi(\mathbf{x}, t) = 1 - \frac{\int_H \mu(t, \xi) dV}{\int_H dV} \quad (12)$$

where φ denotes local damage which is the ratio of the broken bonds to the total bonds that are connected to a material point. The location of the crack tip at any time is defined as the maximum x coordinate of the nodes with the local damage value, φ , exceeding a threshold value of φ_0 . The threshold of local damage, φ_0 , is taken to be 0.3 as suggested by Silling and Askari (2005).

The energy to break all the bonds per unit fracture area is equal to the mode-II fracture toughness, G_{II} , of the material as illustrated in Fig. 3. The relation between the mode-II fracture toughness, G_{II} , and the critical value of the micropotential, ω_c , can be written as (Silling and Askari 2005)

$$G_{II} = \int_0^\delta \int_z^\delta \int_0^{\cos^{-1}(z/\xi)} \omega_c \xi^2 \sin(\theta) d\theta d\xi dz, \quad (13)$$

where the micropotential of a single bond at failure, ω_c , can be written as

$$\omega_c = \frac{c s_c^2 \xi}{2}. \quad (14)$$

The angle, θ , and the distance, z , used in Eq. (13) are illustrated in Fig. 3 and the bond constant c is given in Eq. (9). The critical stretch, s_c , can then be calculated by substituting Eq. (14) in Eq. (13) as

$$s_c = \sqrt{\frac{20 \pi t G_{II}}{9 E \delta^2}}. \quad (15)$$

2.2 Implementation of peridynamic theory in FEA

In PD theory, a continuum body is discretized using material (mass) points that are connected with bonds. The force interactions between these points occur through the bonds. Although PD is considered as one of the “mesh-free” methods, the bond interactions can be represented using a finite element mesh that consists of linear springs or truss elements for bond interactions and concentrated mass for the material points in FEA

codes. Recall that the unit of the Peridynamic force density, f , in Eq. (7) is force per volume squared. The total force in a PD bond, F_{PD} , can then be written as

$$F_{PD} = fV^2 = c_s V^2, \quad (16)$$

where V denotes the volume of a material point and can be calculated as

$$V = (\Delta x)^2 t \quad (17)$$

where x is the PD grid size.

The total force in a truss element, F_{tr} , can be represented in ABAQUS using its geometrical and mechanical properties as

$$F_t = \frac{E_t A_t}{L_t} \Delta L_t, \quad (18)$$

where E_t , A_t , L_t and ΔL_t denote the elastic modulus, the cross sectional area, the initial length and the elongation of the element. Please note that $\Delta L_t/L_t$ is equal to the stretch value, s , as given in Eq. (6). In order to model the stiffness of the PD bonds in FEA software correctly, the total force in a truss element, F_t , should be equal to the peridynamic force, F_{PD} , as defined in Eq. (16). By preserving the conventional units, the following relations can be written:

$$E_t = c \Delta x^4, \quad (19a)$$

$$A_t = t^2. \quad (19b)$$

The density of the truss elements are set to zero since the mass distribution of the continuum is defined via lumped mass, m , at the material points as

$$m = \rho V. \quad (20)$$

In this study, PD bonds are implemented in the commercial finite element code ABAQUS using truss elements as in Macek and Silling (2007). Truss elements (T3D2) in ABAQUS are generated by linking each node to its neighbouring nodes within the horizon using a MATLAB script. The solutions are performed using the explicit solution scheme in ABAQUS software (Abaqus 2014).

3 PD simulation of shear impact in homogeneous media

Shear impact in homogenous linear elastic materials is first investigated experimentally by Rosakis et al. (1999). They conducted experiments on intersonic mode-II crack growth in isotropic solids subjected

to impact shear loading using a specimen prepared by bonding two Homalite plates. In order to obtain homogenous elastic properties and a weak interface, these blocks were bonded together with an adhesive having similar elastic modulus to Homalite. The purpose was to obtain pure mode-II behaviour of crack propagation along the weak plane. Later, Needleman (1999) and Daphalapurkar et al. (2007) conducted numerical studies of the problem considered by Rosakis et al. (1999) using cohesive surface decohesion formulation and generalized interpolation method respectively. In the current study, we investigate the same problem considered by Rosakis et al. (1999) using bond-based PD for the same specimen and the same loading conditions.

3.1 Problem description

There are two different problems considered in this study. First, elastodynamic studies has been carried out for the unnotched Homalite specimen shown in Fig. 4a. Then, dynamic propagation of interface crack between bonded Homalite blocks as shown in Fig. 4b is investigated. Both specimens have a length $L = 150$ mm and a width $w = 75$ mm. The bonded homalite blocks have an initial sharp notch of $a_0 = 25$ mm as illustrated in Fig. 4b. The impact loading is introduced with a length of $b = 50$ mm on the edge where $x = 0$ and $-50 \leq y \leq 0$ using a pulse speed, V_i , given by:

$$V_i(t) = \begin{cases} \frac{V_{it}}{t_r}, & 0 \leq t < t_r \\ V_i, & t_r \leq t < t_r + t_p \\ \frac{V_i(-t+t_r+t_p+t_s)}{t_s}, & t_r + t_p \leq t \leq t_r + t_p + t_s \\ 0, & t_r + t_p + t_s < t < t_f \end{cases} \quad (21)$$

where t_r , t_s , t_p and t_f denote rise time, step-down time, pulse time and termination time respectively as illustrated in Fig. 5. The body is at rest at $t = 0$.

3.2 Homalite material properties

The elastic modulus and the Poisson's ratio of Homalite are $E = 5.2$ GPa and $\nu = 1/3$ respectively. The mass density of the Homalite is $\rho = 1246$ kg/m³. Dilational, c_D , Rayleigh, c_R , and shear, c_S , wave speeds of Homalite can be calculated for plane stress conditions

Fig. 1 Geometry of the material points in the deformed and undeformed configurations

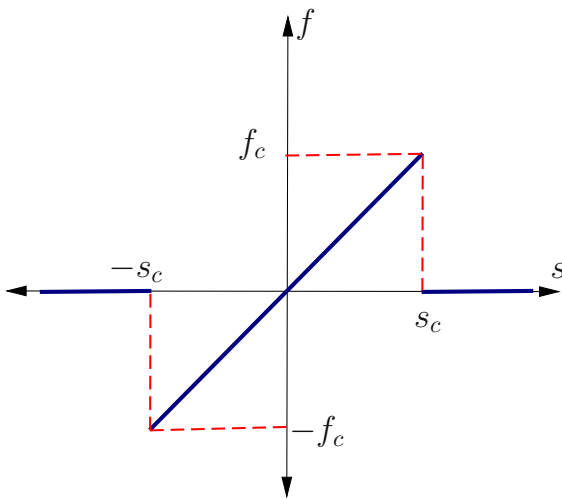
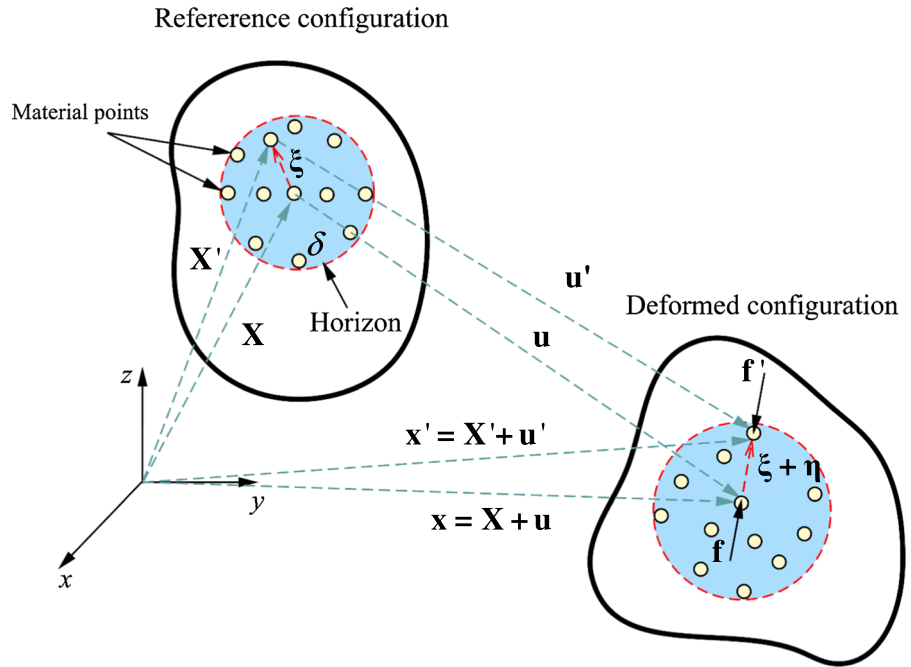


Fig. 2 Modified PMB material model for linear elastic brittle materials

using the following formulas (Freund 1998)

$$c_D = \frac{\kappa + 1}{\kappa - 1} \frac{\mu}{\rho}, \quad (22)$$

$$c_S = \left(\frac{\mu}{\rho} \right)^{1/2}, \quad (23)$$

$$c_R = \frac{0.862 + 1.14\nu}{1 + \nu} c_S, \quad (24)$$

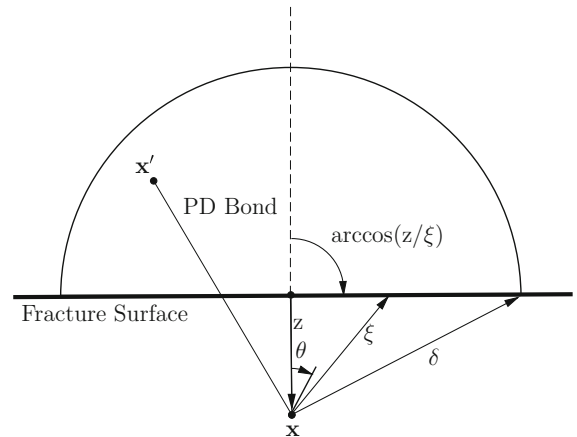


Fig. 3 Calculation of energy to break the PD bonds per unit area

where μ and κ for plane stress condition can be calculated as

$$\mu = \frac{E}{2(1 + \nu)}, \quad \kappa = \frac{3 - \nu}{1 + \nu}. \quad (25)$$

The characteristic wave speeds of Homalite are calculated to be $c_D = 2167$ m/s, $c_S = 1251$ m/s and $c_R = 1165$ m/s. Note that, the crack propagation speed is defined as sub-Rayleigh if the propagation speed is below c_R and as intersonic if the propagation speed is between c_S and c_D . Crack tip speed regime between

Fig. 4 Specimen geometry for **a** single plate used for elastic wave simulations, **b** bonded plates used for crack propagation simulations

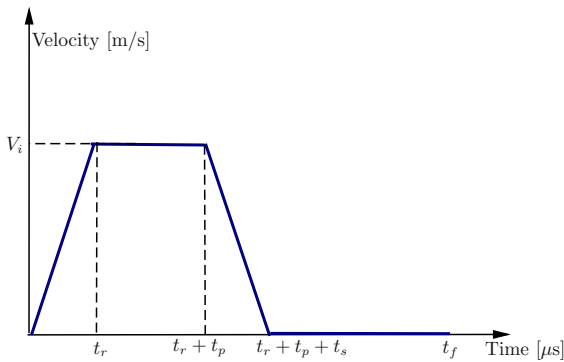
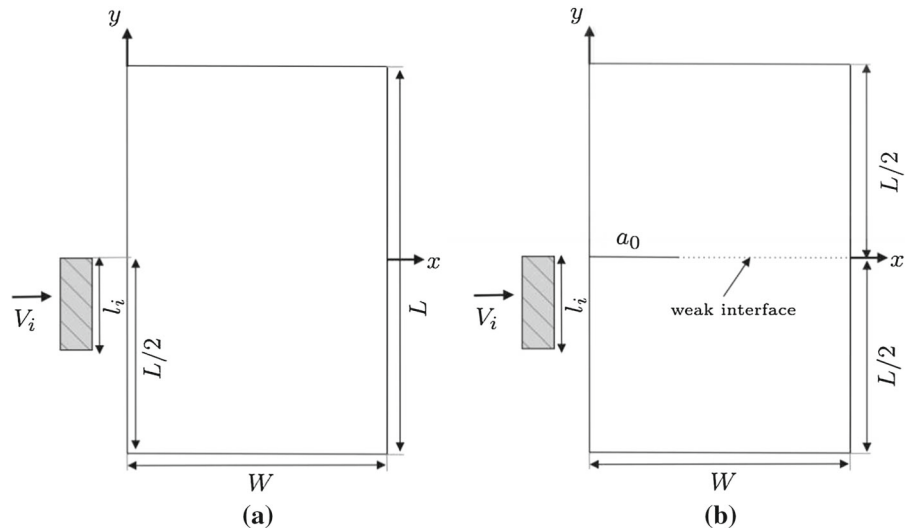


Fig. 5 Time history for impact velocity loading

c_R and c_S is considered to be the forbidden zone since analytical calculations of fracture energy yield negative results in this regime (Broberg 1989).

The PD simulations of dynamic mode-II crack propagation is performed using the methodology described in Sect. 2.2. Peridynamic bonds are generated with mesh sizes ranging from $\Delta x = 0.15$ mm to $\Delta x = 1$ mm by retaining the horizon size $\delta = 3.015\Delta x$ using an in-house MATLAB script. A close-up view of the PD mesh is illustrated in Fig. 6. The simulations are carried out until $t_f = 30$ μ s in an HP-Z420 Desktop workstation having 6 CPUs.

4 Results

PD simulations of projectile impact on the edge of a bonded Homalite specimen are carried out using the

geometry depicted in Fig. 4. Initially, elastodynamic impact simulation on the edge of a homogeneous specimen is carried out to calculate the dilatational wave speed, c_D , and the shear wave speed, c_S , of the Homalite material. Mesh convergence study is performed to obtain the convergent results of crack tip positions and speeds. The interface crack growth is then investigated for various values of impact speed, fracture toughness and pulse time duration. The results are presented in terms of velocity field contours, crack tip positions and velocity histories. Note that, the location of the crack tip at any time is defined as the maximum x coordinate among all nodes that have a higher local damage than $\varphi = 0.3$ as in Silling and Askari (2005).

4.1 Elastodynamic impact simulations

PD simulation of a projectile impact is carried out on an elastic homogeneous Homalite specimen as depicted in Fig. 4a. In this section, only the elastic behaviour of a single Homalite plate is considered by disregarding the damage parameter, μ , given in Eq. (10). The normal impact speed is $V_i = 11$ m/s. The rise time, t_r , and step down time, t_s , are each 0.1 μ s and the pulse time t_p is 10 μ s. The contours of particle velocity magnitude and transverse velocity in the Homalite plate taken at $t = 20$ μ s after impact are shown in Fig. 7a and b, respectively. The distance between the loading wave and the trailing wave in the particle velocity magnitude contour in Fig. 7a is 21 mm which corresponds to the pulse time duration of the applied velocity multiplied

Fig. 6 A close up view of the PD bonds and the material points

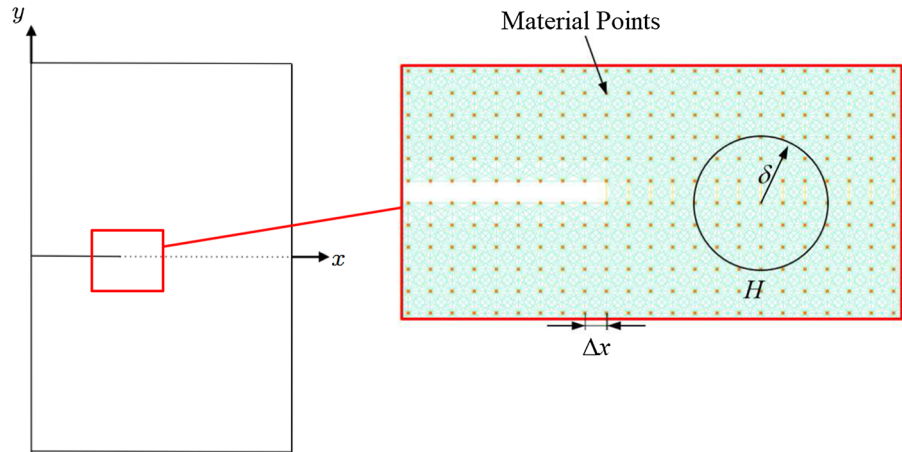
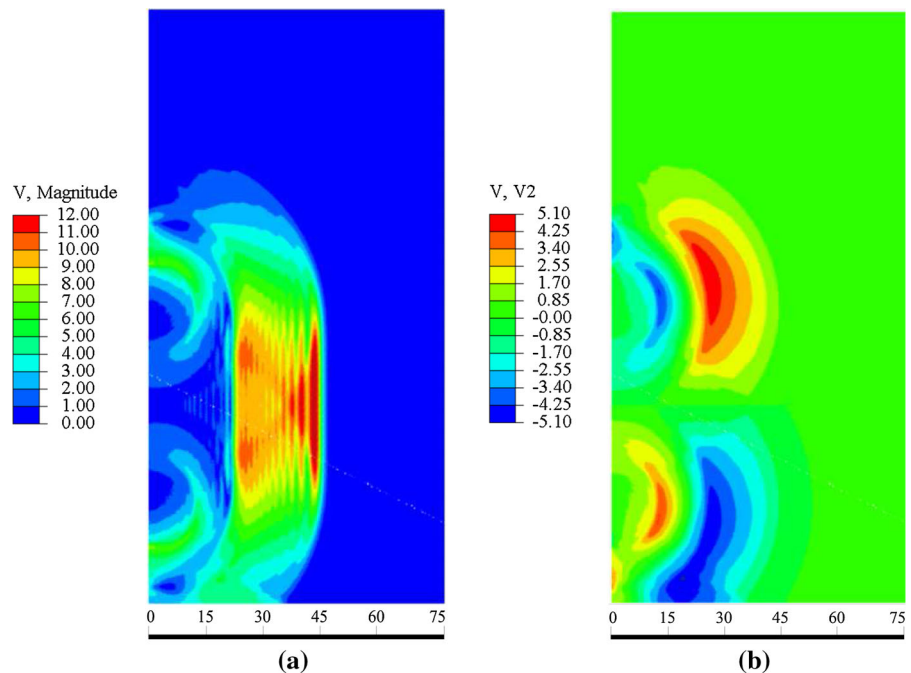


Fig. 7 Contours of **a** the particle velocity magnitude and **b** the transverse particle velocity in the Homalite specimen at $t = 20 \mu\text{s}$



by the longitudinal wave speed. The transverse velocity contour in Fig. 7b indicate that the particle motion is in both $+y$ and $-y$ directions and perpendicular to the direction of wave propagation which is a characteristic feature of the shear wave motion.

The location history of the wave fronts of dilatational and shear waves should be obtained to determine PD estimations of average dilatational wave speed, c_D , and average shear wave speed, c_S . First, we find the location of the material point with the maximum particle velocity behind the dilatational wave front and we scan the material points horizontally in the direction of the

wave. The first material point where the particle velocity is smaller than a threshold value is defined as the dilatational wave front. In our simulations, the threshold value is taken to be 2 m/s. For the shear wave front, we take the x-location of the material point with the highest transverse particle velocity. We apply the same criteria consistently in order to detect the wave fronts with $1 \mu\text{s}$ time intervals.

The locations of the longitudinal wave front and the shear wave front as a function of time are linear as plotted in Fig. 8 and the slopes of the mentioned lines give average longitudinal wave speed, c_D , and the shear

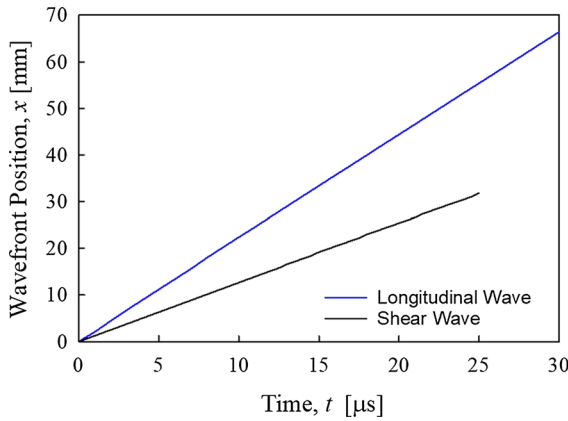


Fig. 8 Wave front position history of **a** the dilatational wave front, **b** the shear wave front

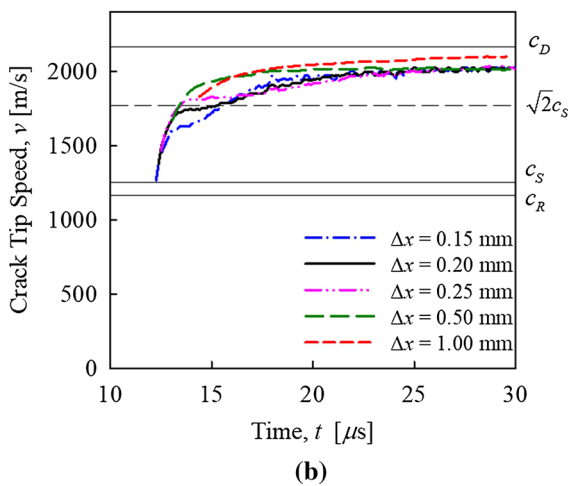
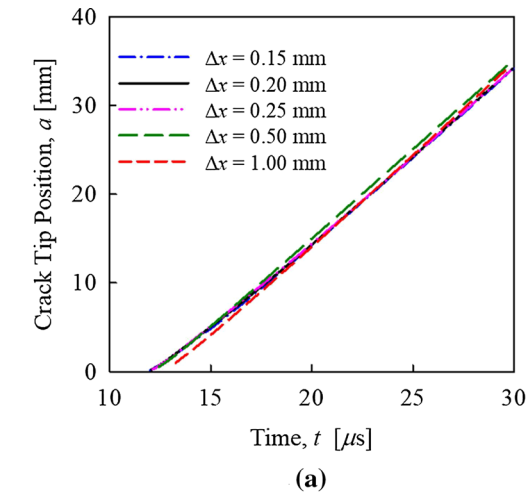


Fig. 9 Time history of **a** the crack tip position, **b** the crack tip speed from the mesh convergence study

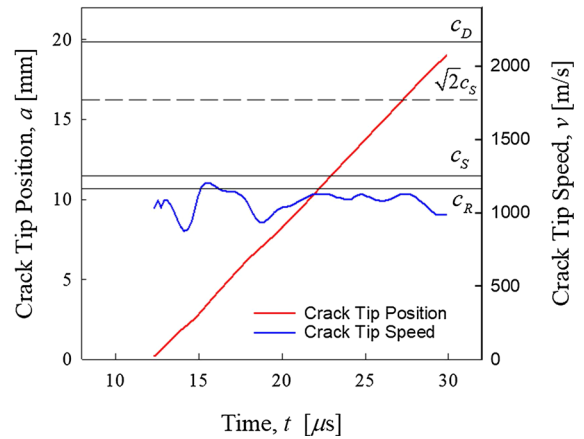


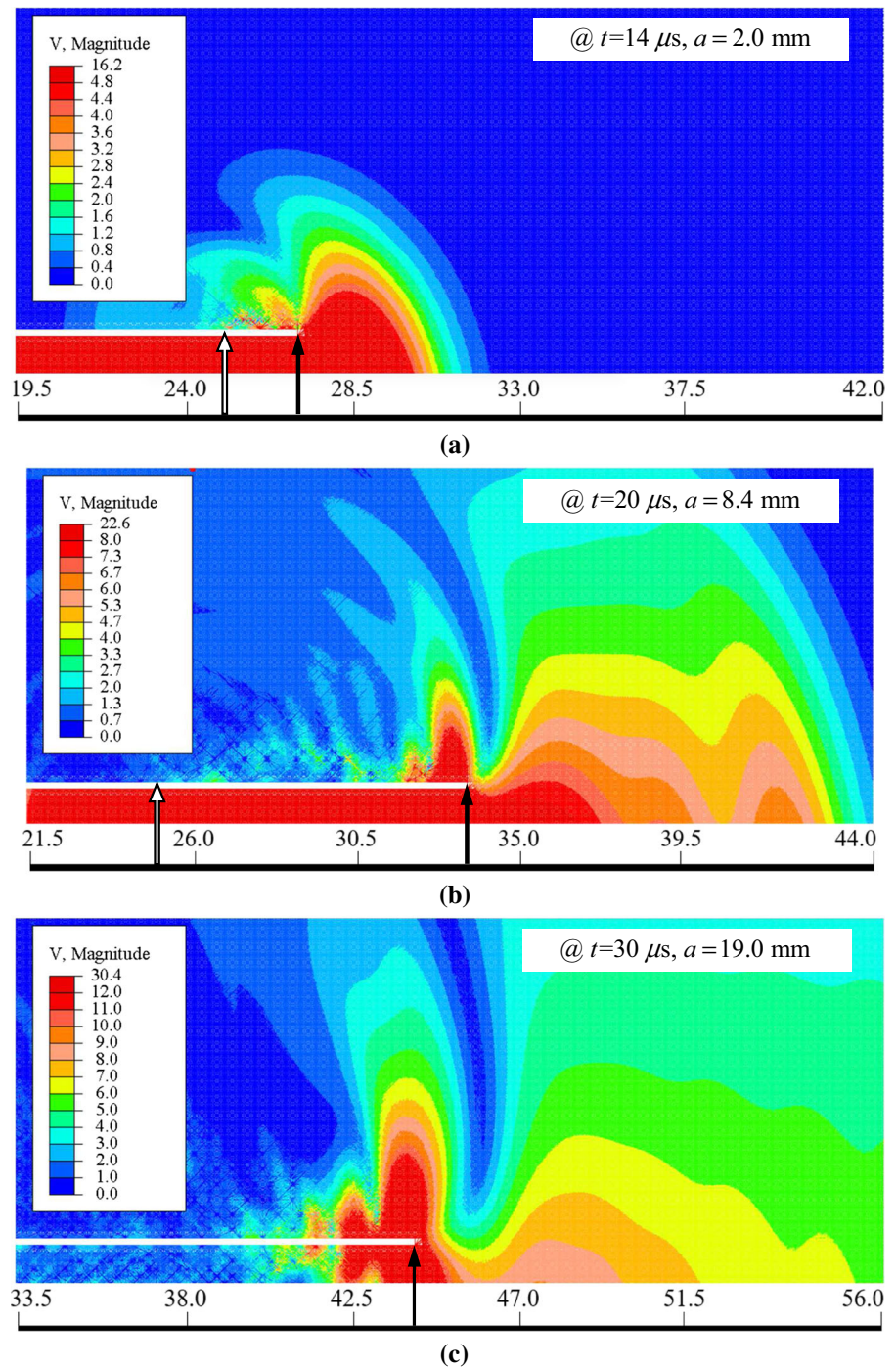
Fig. 10 Time history of the crack tip position and the crack tip speed for $V_i = 10.0$ m/s

wave speed, c_s , as 2210 m/s and 1265 m/s, respectively. The estimated wave speeds are in good agreement with the theoretical values given in Eqs. (22) and (23) with 1.1 % error for the shear wave speed, c_s , and 2.0 % error for the longitudinal wave speed, c_D . These results indicate that PD simulations can accurately capture the elastodynamic behaviour of the Homalite plate.

4.2 Mesh convergence study

PD simulations of dynamic crack propagation in the bonded Homalite plate depicted in Fig. 4b is considered for a mesh convergence study. Damage parameter, μ , in Eq. (10) is activated at the bonding interface which allows the crack to propagate along the interface dynamically. The mesh size, Δx , in Fig. 6 affect the total number of material points that are needed to discretize the PD equations of motion. Thus, the mesh sensitivity of the model is investigated first. Impact speed, V_i , and fracture toughness, G_{II} , are taken as 15 m/s and 1.2 Pa.m in the mesh convergence study. Pulse duration, t_p , is 10 μ s and both the rise time, t_r , and the step down time, t_s , are fixed at 0.1 μ s. Simulations are carried out using mesh sizes $\Delta x = 1.00, 0.5, 0.25, 0.20, 0.15$ mm while fixing the horizon size at $\delta = 3.015\Delta x$. Position of the crack tip, a , which is the distance between the crack tip and initial notch position as a function of time are shown in Fig. 9a for the utilized mesh sizes. It is noticed that crack tip position histories overlap for finer mesh sizes $\Delta x = 0.25, 0.2, 0.1$ mm whereas the coarser mesh sizes $\Delta x = 1.00, 0.5$ mm deviate from

Fig. 11 Particle velocity contours for $V_i = 10.0$ m/s at **a** at $t = 14 \mu\text{s}$, $v_c = 878$ m/s **b** at $t = 20 \mu\text{s}$, $v_c = 1041$ m/s and **c** at $t = 30 \mu\text{s}$, $v_c = 975$ m/s



the others. The speed history of the crack tips are illustrated in Fig. 9b for all mesh sizes in which the crack tip speed at a particular time is obtained by the slope of the crack tip positions using five consecutive points.

As the mesh size decreases, both the initiation time and the initial crack tip speed is found to decrease. For all mesh sizes, the crack growth initiates at speeds above c_R and the crack tip speeds steeply increase to above $\sqrt{2}c_S$. The crack tip speed approaches an inter-

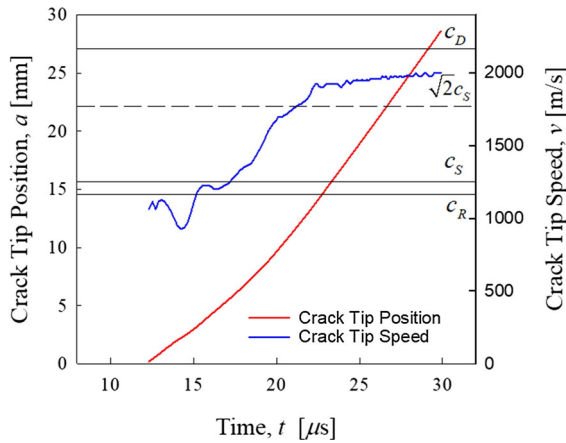


Fig. 12 Time history of the crack tip position and the crack tip speed for $V_i = 10.2$ m/s

sonic steady state speed of $1.62c_S$ with time for all mesh sizes except $\Delta x = 1.00$ mm which continues to rise to $1.67c_S$. The crack tip speed obtained with $\Delta x = 0.50$ mm converges to $1.62c_S$ rapidly whereas the crack tip speeds for smaller mesh sizes reach $1.62c_S$ more gradually around $t = 20 \mu\text{s}$. From these results, it can be concluded that mesh convergence is demonstrated for mesh sizes smaller than $\Delta x = 0.5$ mm. The convergence of the PD results ensure the validity of the applied critical stretch formulation in the present study. Note that the critical stretch in Eq. (15) depends on the horizon size, δ , which is taken as $3.015\Delta x$. Hereinafter, the mesh size $\Delta x = 0.2$ mm is chosen for further investigations in the current study.

4.3 Effect of impact speed, V_i , on dynamic crack propagation

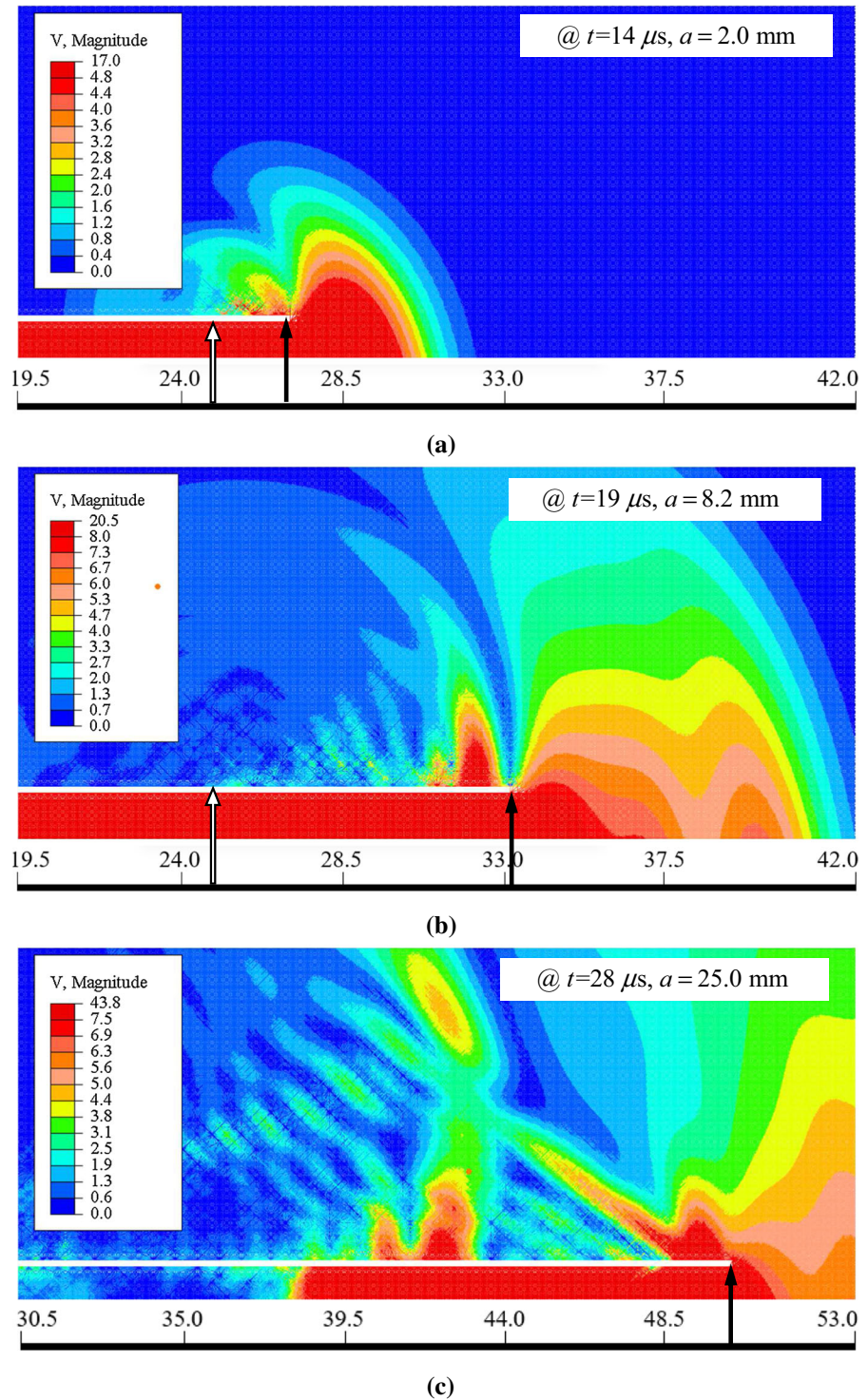
In this section, the different regimes of crack growth is investigated for varying impact speeds at a fixed mode-II fracture toughness value of $G_{II} = 1.2$ Pa.m. The simulations are presented for impact speeds of 10 m/s, 10.2 m/s and 25 m/s, demonstrating sub-Rayleigh, transition from sub-Rayleigh to intersonic and intersonic crack growth regimes. The impact pulse duration is $10 \mu\text{s}$, and the rise and step-down times are $0.1 \mu\text{s}$. Three cases are presented in detail where sustained crack tip speeds are attained at c_R , $1.57c_S$ and $1.66c_S$.

In the first case where the impact speed is $V_i = 10.0$ m/s, the crack propagation starts at $t = 12.3 \mu\text{s}$ in the sub-Rayleigh regime. Figure 10 shows the crack tip

position and crack tip speed history. The crack tip position history is nearly linear and the crack tip speed fluctuates just below the Rayleigh wave speed, c_R , during the entire simulation. The particle velocity magnitude contours around the crack tip at $t = 14.0, 20.0, 30.0 \mu\text{s}$ are illustrated in Fig. 11a–c, respectively. Figure 11a corresponds to $1.67 \mu\text{s}$ after the crack initiation where the crack tip position a is 2.0 mm and the crack tip speed is 878 m/s. The initial notch position is marked with a hollow arrow and the crack tip location is marked with a solid arrow. The crack tip location a at $t = 20 \mu\text{s}$ is 8.4 mm as seen in Fig. 11b and the corresponding crack tip speed is 1132 m/s. The crack tip position a reaches 19.0 mm at $t = 30 \mu\text{s}$ at a crack tip speed of $v_c = 975$ m/s. The particle velocity concentrations around the crack tip forms a smooth round shape in front of the crack tip in Fig. 11 which is a characteristic behaviour for the sub-Rayleigh crack tip speed regime. It is also noticed that the shape of the concentration gets bigger as the crack propagation distance increases.

In the next case, the impact speed, V_i , is increased from 10.0 m/s to 10.2 m/s. In this case, the crack initiation time is $t = 12.3 \mu\text{s}$ which is the same with the previous case. The crack tip position and the crack tip histories are shown in Fig. 12. There is a noticeable change in the slope of the crack tip position history after $t = 17 \mu\text{s}$ which indicates the increase in the crack tip speed. As seen from the crack tip speed plot, the crack propagation starts in the sub-Rayleigh regime and rises to a crack tip speed of $1.57c_S$ at $t = 22.3 \mu\text{s}$ which is then sustained in the rest of the simulation. The particle velocity contours around the crack tip at $t = 14.0 \mu\text{s}$ is shown in Fig. 13a where the crack tip speed is 943 m/s. The crack tip location a is 2.0 mm at $t = 14 \mu\text{s}$ which is the same as the previous case and the distribution of the particle velocity contours are nearly identical at this moment. Particle velocity contours around the crack tip at $t = 19 \mu\text{s}$ is shown in Fig. 13b with a corresponding crack tip location of 8.2 mm and an intersonic speed of 1501 m/s. Due to the transition from the sub-Rayleigh regime to intersonic regime, the shape of the particle velocity contours changes and the concentrations which were around the crack tip before $t = 17 \mu\text{s}$ fall behind the crack tip. The particle velocity contours at $t = 28 \mu\text{s}$ where the crack tip speed has reached a value of $1.57c_S$ is shown in Fig. 13c. The corresponding crack tip location and the crack tip speed is $a = 25.0$ mm and $v_c = 1959$ m/s, respectively. At this moment, the discontinuity lines, shear Mach waves, originating

Fig. 13 Particle velocity contours for $V_i = 10.2$ m/s at **a** at $t = 14 \mu\text{s}$, $v_c = 943$ m/s, **b** at $t = 19 \mu\text{s}$, $v_c = 1501$ m/s and **c** at $t = 28 \mu\text{s}$, $v_c = 1959$ m/s



from the crack tip are seen in Fig. 13c as an indication of the intersonic speed regime. The inclination of shear Mach waves with the horizontal lines is illustrated in

Fig. 14, called the shear Mach angle, β , and can be calculated as (Rosakis 2002)

$$\beta = \sin^{-1} \left(\frac{c_S}{v_c} \right), \quad (26)$$

where v_c is the crack propagation speed. The inclination of the shear Mach waves in Fig. 13b is measured approximately to be 40.5° which agrees with the value of $\beta = 39.8^\circ$ obtained using Eq. (26) where instantaneous crack tip speed from the snapshots are used. The lowest sustained intersonic crack speed for the interface toughness of $G_{II} = 1.2$ Pa.m is attained at $1.57c_S$ for an impact speed of $V_i = 10.2$ m/s based on the PD simulations.

Next, the crack tip position and crack tip speed with respect to time is illustrated in Fig. 15 for the case of a higher impact speed of $V_i = 25.0$ m/s. The crack tip position history is linear after the initiation part. The crack initiates at $t = 11.8 \mu\text{s}$ in the intersonic regime and a sustained intersonic crack tip speed is reached at $1.65c_S$ rapidly. The particle velocity contours are shown in Fig. 16a at $t = 14.0 \mu\text{s}$ where the crack tip location is $a = 4.4$ mm and the crack tip speed is v_c is 2030 m/s. Note that, the crack tip position was $a = 2.0$ mm in the previous cases at the same time. As a result of intersonic crack propagation, the position of the crack tip is closer to the dilatational wave front which is in light green colour compared to other cases. The crack tip position is $a = 16.8$ mm at $t = 20.0 \mu\text{s}$. The crack tip positions are $a = 16.8$ mm and 37.2 mm at $t = 20.0 \mu\text{s}$ and $t = 30.0 \mu\text{s}$, respectively in Figs. 16b, c. The corresponding crack tip speeds are 2060 m/s and 2083 m/s. The velocity field patterns in Fig. 16 contain shear Mach waves which are characteristic behaviour of the intersonic crack propagation. The theoretical value of the shear Mach angle, β , around a crack propagating at $1.65c_S$ is calculated as 37.5° and the inclination of the shear Mach waves in Fig. 16c are in good agreement with this value using the calculated crack tip speeds.

4.4 Effect of fracture toughness, G_{II} , on dynamic crack propagation

The effect of fracture toughness on the dynamic crack growth is investigated for a fixed impact speed of $V_i = 15$ m/s. The pulse width is $t_p = 10 \mu\text{s}$ and rise t_r and step down t_s times are $0.1 \mu\text{s}$ as before. Three fracture toughness values that lead to three distinct regimes are chosen for detailed discussion. Cases I and II are discussed together in which the crack tip speeds during the analysis are entirely in the sub-Rayleigh and the inter-

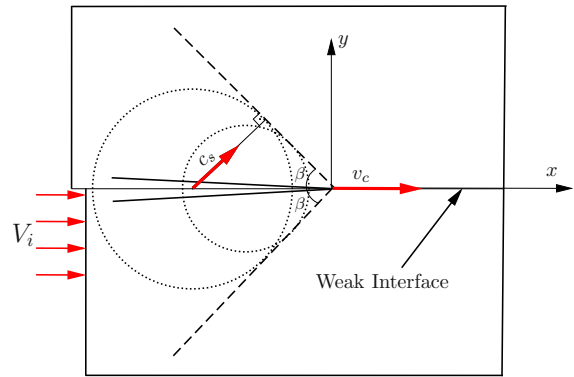


Fig. 14 An illustration of the shear Mach waves, inclined at a Mach angle, β originating from a mode-II crack tip moving at a speed v_c

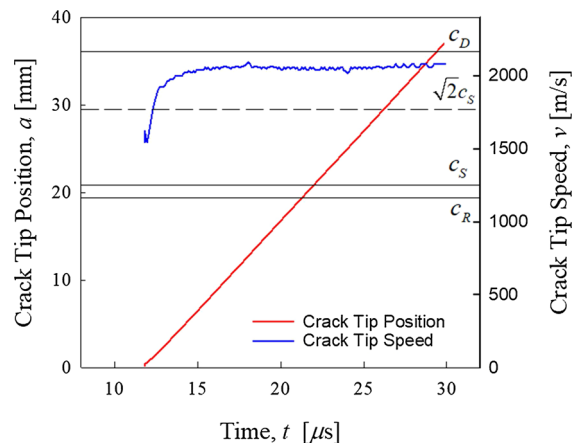
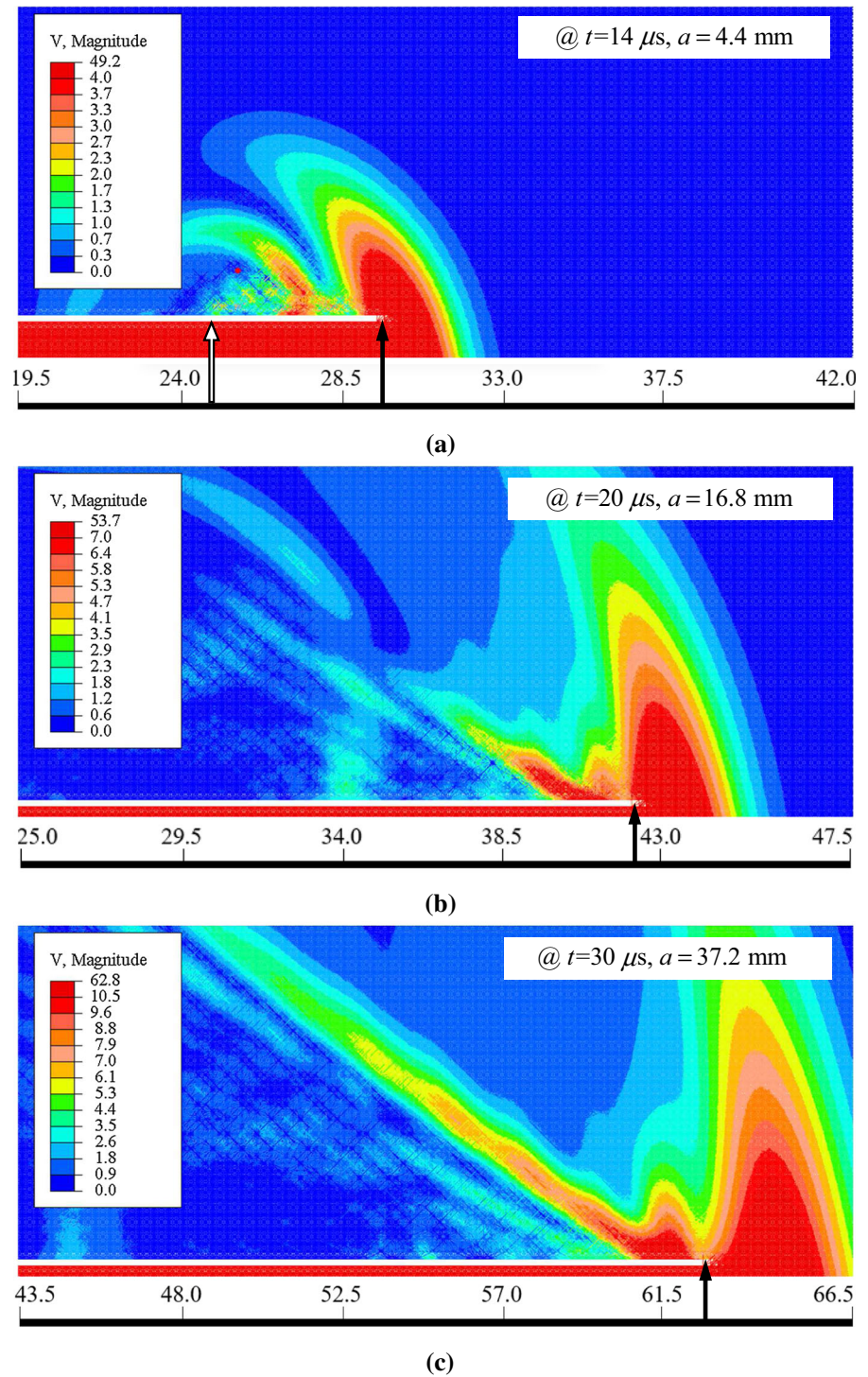


Fig. 15 Time history of the crack tip position and the crack tip speed for $V_i = 25.0$ m/s

sonic regimes, respectively. The third case discussed is the transition case where the crack tip speed jumps from the sub-Rayleigh regime to intersonic regime and sustains at $1.57c_S$. For all cases, the crack tip position and speed histories are presented in addition to particle velocity contours.

In the first two cases, the fracture toughness values are $G_{II} = 3$ Pa.m and 0.6 Pa.m at a constant impact speed of $V_i = 10$ m/s for which the crack tip position and speed histories are depicted in Fig. 17a and b, respectively. The steady state crack tip speed is just below c_R for $G_{II} = 3$ Pa.m and the initiation time is $t = 12.4 \mu\text{s}$. For $G_{II} = 0.6$ Pa.m, the crack tip speed rises to 2057 m/s ($1.64c_S$) after initiating at $t = 11.9 \mu\text{s}$ with an initiation speed of 1500 m/s. In both cases,

Fig. 16 Particle velocity contours for $V_i = 25.0$ m/s at **a** at $t = 14 \mu\text{s}$, $v_c = 2030$ m/s, **b** at $t = 20 \mu\text{s}$, $v_c = 2060$ m/s and **c** at $t = 30 \mu\text{s}$, $v_c = 2083$ m/s



the crack tip speeds remain in the speed regimes that the crack has initiated.

The particle velocity contours of both cases at steady state are depicted in Fig. 18. For the sub-Rayleigh case, the particle velocity distribution at $t = 25 \mu\text{s}$ is illustrated in Fig. 18a where the crack tip location $a = 13.4 \text{ mm}$ and the crack tip speed is $v_c = 1068 \text{ m/s}$. Note that, the velocity contours in Fig. 18a is very similar to the contours in Fig. 11c in Sect. 4.3 which is obtained for $V_i = 10 \text{ m/s}$ and $G_{II} = 1.2 \text{ Pa}\cdot\text{m}$ and under the same pulse duration. For the intersonic case ($G_{II} = 3 \text{ Pa}\cdot\text{m}$), the particle velocity fringes at $t = 28 \mu\text{s}$ is shown in Fig. 18b where the crack tip location $a = 32.2 \text{ mm}$ and the crack tip speed is $v_c = 2057 \text{ m/s}$ ($1.64c_s$). The shear Mach waves that are observed in Fig. 18b are oriented at $\beta = 38.6^\circ$ which is identical to the theoretical value of β . In Sect. 4.3, we obtained nearly identical steady state speed ($1.65c_s$) for the interface crack with $V_i = 25.0 \text{ m/s}$ and $G_{II} = 1.2 \text{ Pa}\cdot\text{m}$ as shown in Figs. 16a-c. In both cases pure intersonic crack propagation is observed around the steady state crack tip speed of $V_i = 1.6c_s$.

As the third case, we investigate the dynamic crack growth of an interface crack with a fracture toughness of $G_{II} = 2.5 \text{ Pa}\cdot\text{m}$ which illustrates the transition from sub-Rayleigh to intersonic crack growth. The crack tip position and speed histories are illustrated in Fig. 19. The crack initiates in the sub-Rayleigh regime at $t = 12.3 \mu\text{s}$ which is very close to that of sub-Rayleigh case ($G_{II} = 3 \text{ Pa}\cdot\text{m}$) which initiated at $t = 12.4 \mu\text{s}$. The crack tip speed of the transition case increases and jumps to intersonic speed regime around $t = 15.5 \mu\text{s}$ and reaches a steady state intersonic speed of $v_c = 1.57c_s$ after $t = 21 \mu\text{s}$. The particle velocity contours at $t = 14 \mu\text{s}$ is shown in Fig. 20a where the crack tip location $a = 2.0 \text{ mm}$ and the crack tip speed is $v_c = 1041 \text{ m/s}$. The particle velocity distributions of the transition case at $t = 14 \mu\text{s}$ is similar to previous sub-Rayleigh cases in Figs. 11a, 13a. After $t > 14 \mu\text{s}$, the crack tip speed starts to increase with time as seen in Fig. 19. The particle velocity contours at $t = 17 \mu\text{s}$ is shown in Fig. 20b where the crack tip location is $a = 5.6 \text{ mm}$ and crack tip speed is $v_c = 1400 \text{ m/s}$. Note that, the crack tip speed is in transition from sub-Rayleigh to intersonic at that moment and that there are two regions of particle velocity concentrations. Later, these two regions coalesce and propagate at intersonic speed of $v_c = 1.57c_s$ as shown in Fig. 20c which is a snapshot taken at $t = 28 \mu\text{s}$ and $a = 25.8 \text{ mm}$.

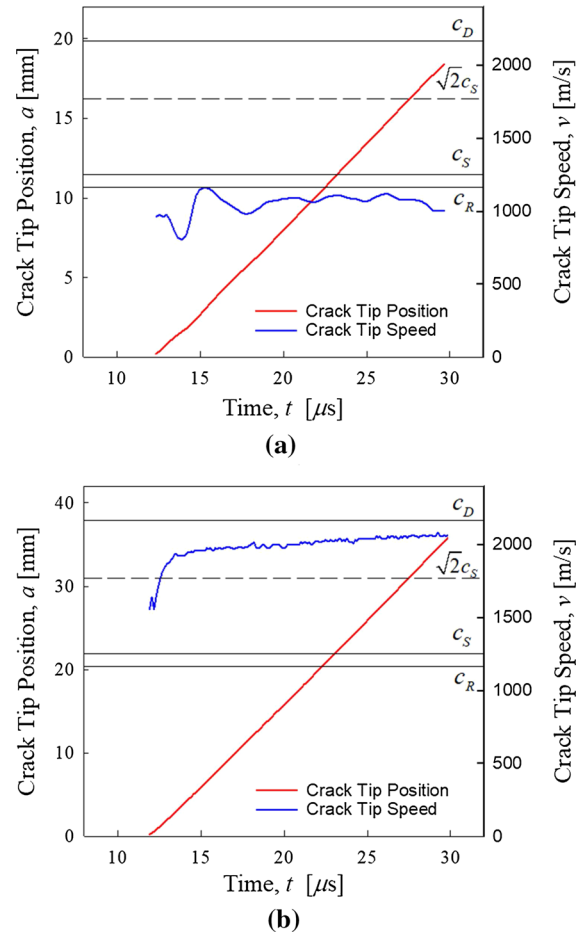


Fig. 17 Time history of the crack tip position and the crack tip speed for **a** $G_{II} = 3.0 \text{ Pa}\cdot\text{m}$, **b** $G_{II} = 0.6 \text{ Pa}\cdot\text{m}$

Corresponding shear Mach waves emanating from the crack tip are also observed with an angle of $\beta = 39.8^\circ$. The traces of the round shaped concentration behind the crack tip can also be seen at position $x = 42 \text{ mm}$ in the particle velocity contours at Fig. 20c that is seen to propagate at the Rayleigh wave speed. This trace is the signal of the transition from the sub-Rayleigh speed regime to the intersonic speed regime. Similar concentrations is also seen in the particle velocity contours of the case $V_i = 10.2 \text{ m/s}$ in Fig. 13b, c in Sect. 4.3. We observe that, this signal also remains in the contours and travels with the speed of c_R .

One important aspect of this transition can be alluded from the crack tip opening profiles. The crack tip opening displacements for the three different cases are given in Fig. 21. Opening displacements in Fig. 21a

Fig. 18 Particle velocity contours for **a** $G_{II} = 3.0$ Pa.m at $t = 25 \mu\text{s}$, $v_c = 1068$ m/s, **b** $G_{II} = 0.6$ Pa.m $t = 28 \mu\text{s}$, $v_c = 2057$ m/s

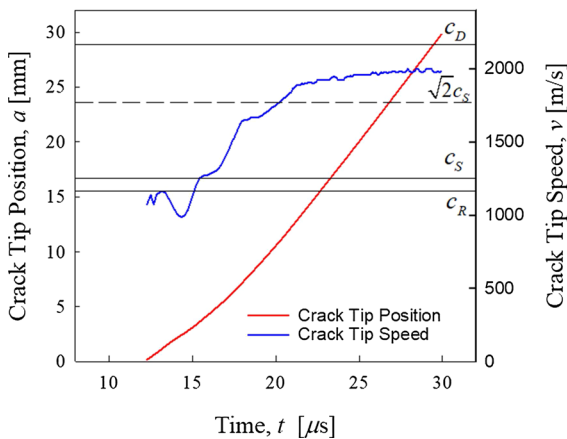
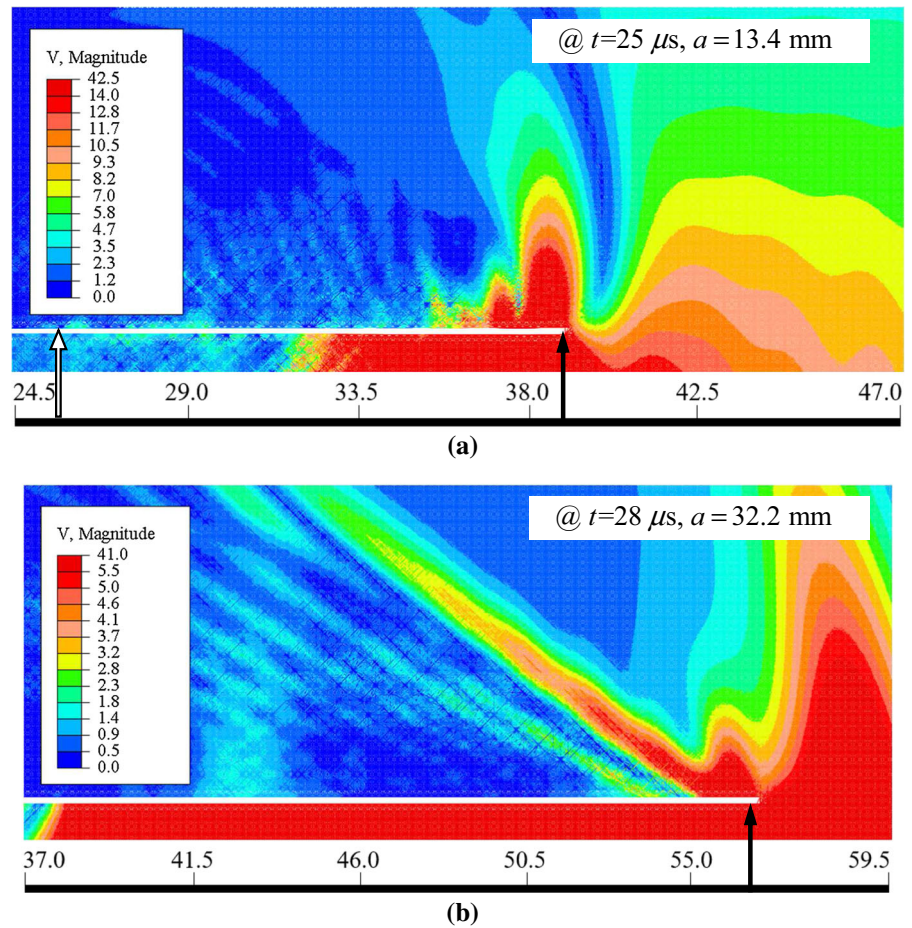
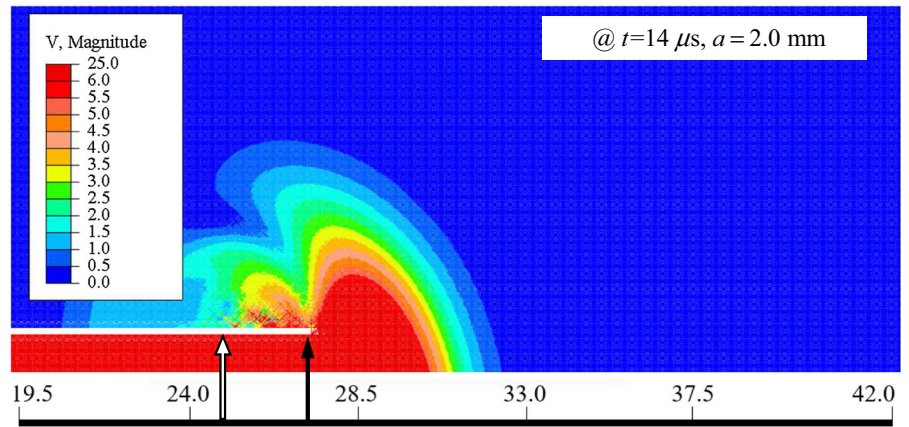


Fig. 19 Time history of the crack tip position and the crack tip speed for $G_{II} = 2.5$ Pa.m

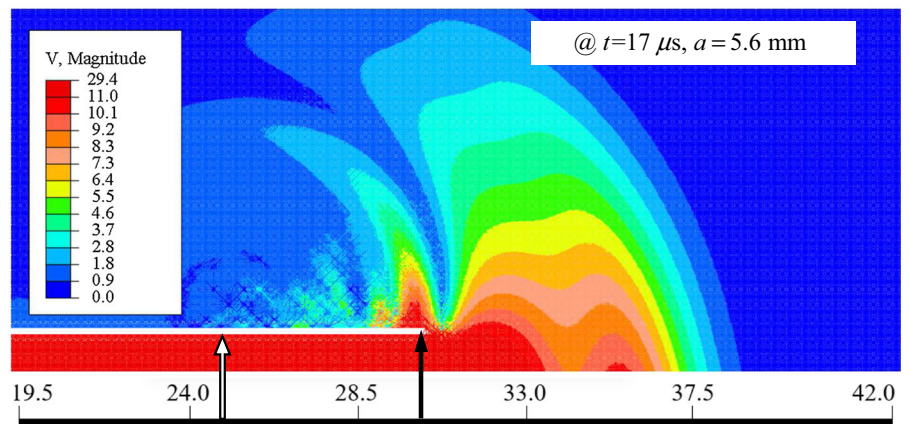
and b indicate that there is no contact behind the crack tip for sub-Rayleigh and intersonic cases. The slope of

the crack tip opening displacement behind the crack tip for $G_{II} = 0.6$ Pa.m (purely intersonic case) have a steeper slope than that for $G_{II} = 3$ Pa.m (purely sub-Rayleigh case). In addition, the opening displacement behind the crack tip rises from the crack tip. However, for the transition case, we can see an indication of the transition from the sub-Rayleigh regime to the intersonic speed regime in Fig. 20c. The crack tip speed is in the sub-Rayleigh regime at $t = 13 \mu\text{s}$ and there is no contact behind the crack tip at that moment as seen in Fig. 21c. After $t = 14 \mu\text{s}$, there is a steep increase in the crack tip speed as seen in the crack speed history plot of the current case in Fig. 19. We observe a closure effect behind the crack tip in crack tip opening displacements at $t = 14, 15$ and $17 \mu\text{s}$. Note that, the crack tip speed rapidly increases in this time interval and accelerates to the steady state speed of $1.57c_S$. Such contact points behind the crack tip is not seen in the crack tip opening displacements of purely sub-Rayleigh and purely

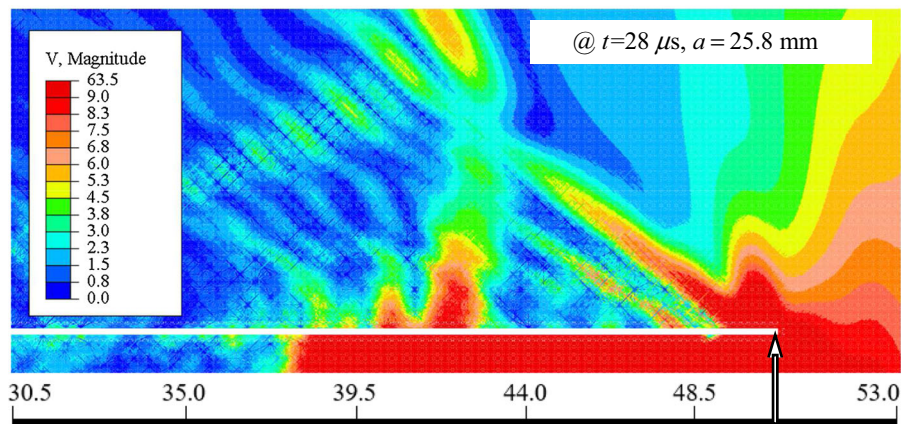
Fig. 20 Particle velocity contours for $G_{II} = 2.5$ Pa.m at **a** at $t = 14 \mu\text{s}$, $v_c = 1041$ m/s, **b** at $t = 17 \mu\text{s}$, $v_c = 1400$ m/s and **c** at $t = 28 \mu\text{s}$, $v_c = 1984$ m/s



(a)



(b)



(c)

inter-sonic cases. In the current case, the contact point behind the crack tip vanishes for times after $t = 21 \mu\text{s}$, when the crack tip speed reaches the sustained inter-sonic speed.

5 Discussion

In this section, a parametric study is conducted to study the effect of impact speed and fracture toughness on the crack tip speeds. In addition, the effect of pulse width and comparison with the literature are presented. In the plots, all the speeds are normalized with respect to the shear wave speed, c_S , and denoted with asterisk.

In Fig. 22a, the history of the normalized crack tip speeds is shown for impact speeds between 10 m/s and 50 m/s at a fixed fracture toughness of $G_{II} = 1.2 \text{ Pa.m}$. For low impact speeds up to 10.1 m/s, the crack tip speed is sub-Rayleigh throughout the simulation. At an incrementally higher impact speed of 10.2 m/s, the crack initiates at a sub-Rayleigh speed and climbs to an inter-sonic speed of $1.57c_S$ gradually demonstrating a sub-Rayleigh to inter-sonic speed transition behaviour. At higher impact speeds of 11–25 m/s, the crack initiates and propagates in the inter-sonic speed regime throughout the simulation, eventually reaching a sustained crack tip speed between $1.57c_S$ and $1.66c_S$. With increasing impact speed, the inter-sonic crack tip speed at initiation is found to increase and the rise time of the crack tip speed to decrease. At an impact speed of 11 m/s, the crack initiates at a speed just above c_R and gradually rises to the sustained crack tip speed of $1.57c_S$. As the impact speed is increased, the initiation speed increases from $1.25c_S$ to $1.5c_S$ rising steeply to the sustained crack tip speed values that vary from $1.57c_S$ to c_D with a sustained crack tip speed of c_D attained for an impact speed of 50 m/s.

In Fig. 22b, the history of the normalized crack tip speeds is shown for fracture toughness values ranging from 6 Pa.m to 0.1 Pa.m, at a fixed impact speed of $V_i = 15 \text{ m/s}$. Similar crack tip speed regimes are observed as before, this time as a function of the fracture toughness parameter. As the fracture toughness is reduced from 6 Pa.m to 3 Pa.m, the crack propagation takes place in the sub-Rayleigh regime throughout the simulation, with the initial crack tip speed increasing with decreasing toughness but not passing c_R . As the fracture toughness is further reduced to 2.5 Pa.m, the crack initiates at a sub-Rayleigh speed just

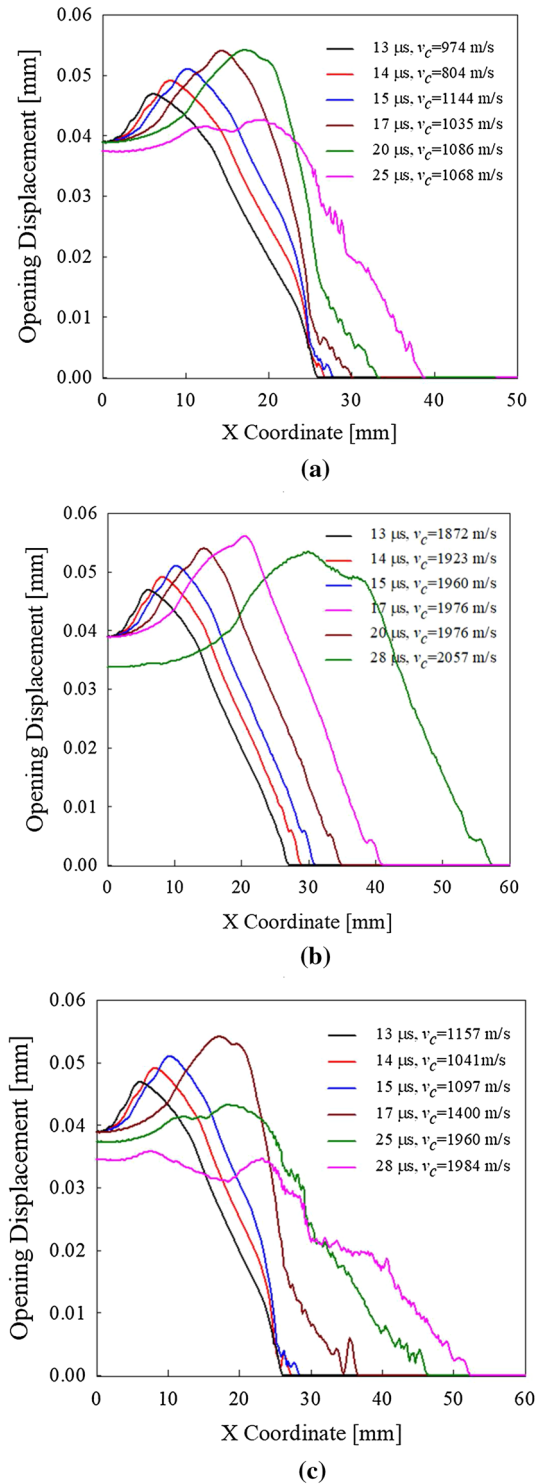


Fig. 21 Variation of crack opening displacement with time for **a** for $G_{II} = 3.0 \text{ Pa.m}$, **b** for $G_{II} = 0.6 \text{ Pa.m}$ and **c** for $G_{II} = 2.5 \text{ Pa.m}$

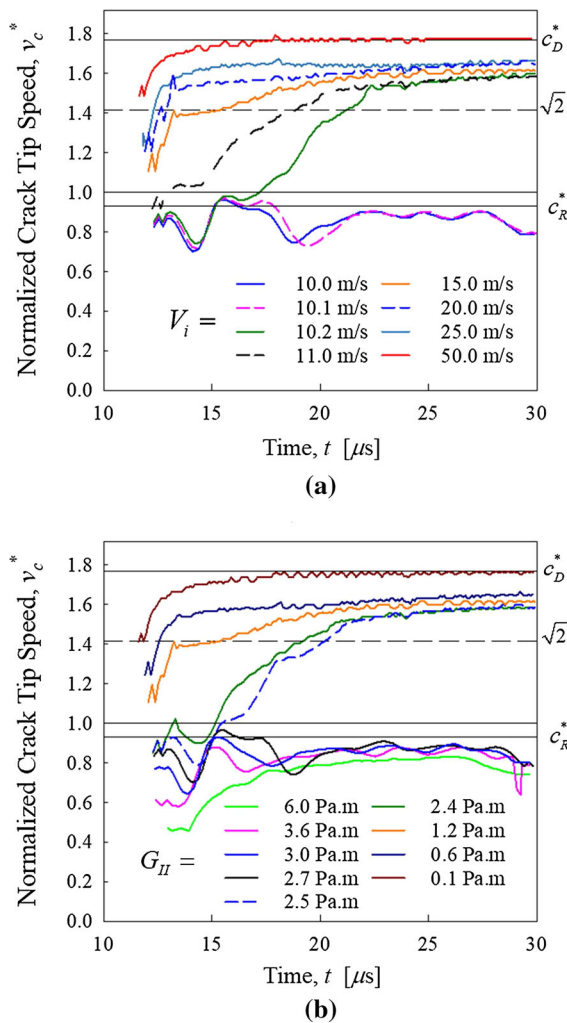


Fig. 22 Time history of the crack tip speed for **a** fixed fracture toughness and **b** fixed impact speed conditions

below c_R , propagating at a speed near c_R and then gradually increasing to an intersonic speed of $1.57c_S$ demonstrating sub-Rayleigh to intersonic speed transition behaviour. A similar transition behaviour is also observed for a lower toughness value of $2.4 \text{ Pa}\cdot\text{m}$. As the toughness values are further decreased to 1.2 and $0.6 \text{ Pa}\cdot\text{m}$, the crack tip speed rapidly increases to the sustained crack tip speeds above $1.57c_S$. As the toughness values are further decreased from $1.2 \text{ Pa}\cdot\text{m}$ to $0.1 \text{ Pa}\cdot\text{m}$, the crack tip speed initiates at increasing intersonic speeds, rapidly reaching the sustained crack tip speed values between $1.57c_S$ and c_D . At the lowest toughness value of $0.1 \text{ Pa}\cdot\text{m}$, the crack tip speed reaches a steady speed value of c_D .

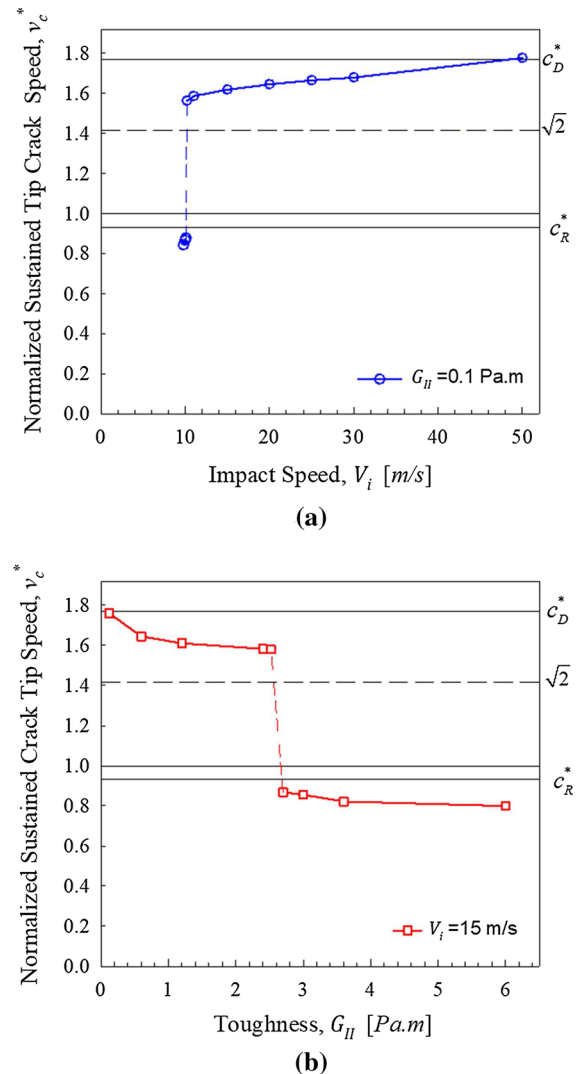


Fig. 23 Variation of the normalized crack tip speed for **a** fixed fracture toughness, $G_{II} = 1.2 \text{ Pa}\cdot\text{m}$ and **b** fixed impact speed, $V_i = 15 \text{ m/s}$

In both Fig. 22a and b, it can be seen that the transition time of the crack tip speed from a speed just below c_R to the sustained intersonic crack tip speed of $1.57c_S$ occurs gradually over a time duration of $10 \mu\text{s}$. This gradual rise of the crack tip speed in our simulations is in good agreement with the experimental results of Rosakis et al. (1999) and Samudrala et al. (2002) where they observed a gradual transition to the intersonic speed in approximately $10\text{--}15 \mu\text{s}$. On the other hand, the numerical simulations of Needleman (1999) and Daphalapurkar et al. (2007) obtained rapid tran-

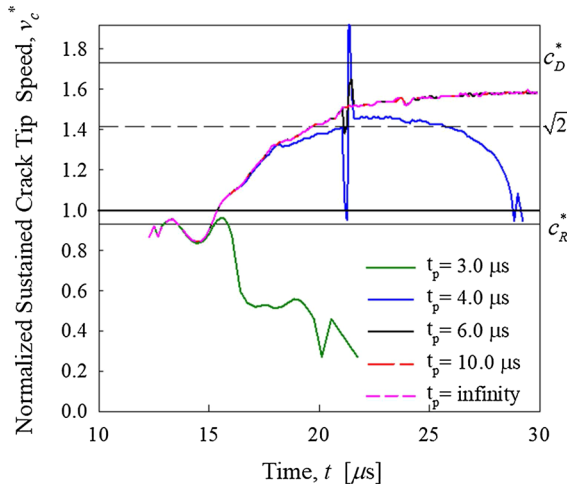


Fig. 24 Variation of the normalized crack tip speed with time for different pulse durations ($G_{II} = 1.2$ Pa.m, $V_i = 10.5$ m/s)

sition to a sustained intersonic crack tip speed above $\sqrt{2}c_S$ in a short time duration of 1–2 μ s.

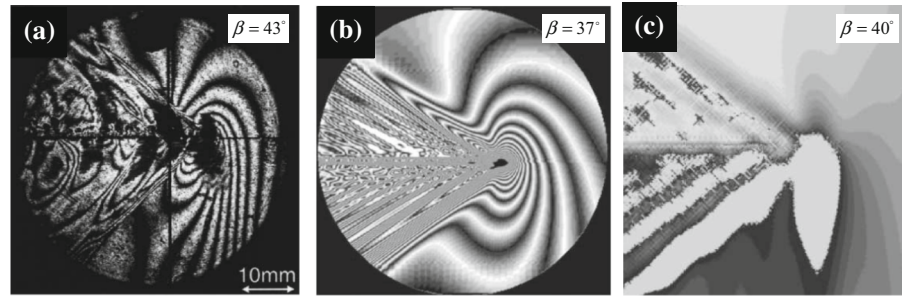
The sustained crack tip speeds obtained for varying impact speeds at fixed $G_{II} = 1.2$ Pa.m and for varying fracture toughness at fixed $V_i = 15$ m/s are summarized in Fig. 23a and b. These plots clearly show the jump of the sustained crack tip speed from sub-Rayleigh to intersonic speeds between $1.57c_S$ and c_D , at a critical velocity of 10.1 m/s (Fig. 23a) or a critical fracture toughness of 2.5 Pa.m (Fig. 23b). Thus, for impact loading under a pulse duration of 10μ s, the sustained crack tip speeds are found to take place in two distinct regimes: the Sub-Rayleigh speed regime below c_R , and the intersonic speed regime between $1.57c_S$ and c_D . Sustained intersonic crack tip speeds below $1.57c_S$ are not observed for impact loading with a pulse duration of 10μ s.

As noted by Burridge et al. (1979), the theoretical value of the lowest stable intersonic crack growth speed is $\sqrt{2}c_S$. However, the lowest stable intersonic crack tip speeds obtained are higher than the theoretical value of the lowest stable intersonic crack tip speed $\sqrt{2}c_S$ in our simulations. Needleman (1999) and Daphalapurkar et al. (2007) also observed minimum sustained intersonic crack tip speeds that are higher than $\sqrt{2}c_S$ consistent with our results, albeit at higher minimum intersonic speeds. Needleman (1999) investigated the dependency of the lowest stable intersonic crack tip speed on the pulse duration and he explored that, the crack tip speed rapidly accelerates to a value greater

than $\sqrt{2}c_S$ and then decrease to $\sqrt{2}c_S$ for a short time period as the pulse width is sufficiently narrow. Abraham and Gao (2000) conducted two-dimensional molecular dynamics simulations of intersonic crack propagation and observed that the sustained crack tip speed dropped from the longitudinal wave speed to $\sqrt{2}c_S$ if the applied pulse was adjusted so that applied loads are completely released as soon as the daughter crack initiated. For higher pulse durations, they obtained sustained crack tip speeds around c_D (Abraham and Gao 2000). In our study, we also investigated the effect of pulse duration on the behaviour of the dynamic shear crack propagation. For this purpose, the interface toughness, G_{II} , and the impact speed, V_i , are fixed at 1.2 Pa.m and 10.5 m/s, respectively for the pulse time durations of $t_p = \infty, 10, 6, 4$ and 3μ s. The crack tip speed histories are shown in Fig. 24. The crack tip speeds overlap initially for all pulse durations. The crack tip speed rapidly decelerates after $t = 17 \mu$ s for the pulse time duration of $t_p = 3 \mu$ s. The crack tip speed accelerates for the other pulse time durations until $t = 18 \mu$ s. After this moment, the crack tip speed for $t_p = 4 \mu$ s starts to deviate from the others. It accelerates to 1830 m/s and then decelerates to $\sqrt{2}c_S$ (1769 m/s) at $t = 25 \mu$ s. The intersonic crack tip speed of $\sqrt{2}c_S$ is maintained between times $t = 25.0 \mu$ s and 26.8μ s. After $t = 26.8 \mu$ s, the crack speed rapidly decreases to c_R . This behaviour is consistent with the theoretical studies of Burridge et al. (1979) as they conclude that crack tip speeds between c_S and $\sqrt{2}c_S$ are unstable. Our simulation results with varying pulse durations in Fig. 24 suggest that it is possible to attain the propagation speed $\sqrt{2}c_S$ for shorter pulse times as also concluded by Needleman (1999); Abraham and Gao (2000).

Finally, we compare the general shape of the shear Mach waves with the shear impact experiments of Rosakis et al. (1999) and numerical study of Daphalapurkar et al. (2007) for the lowest sustained intersonic speeds. The experimental shear stress pattern is illustrated in Fig. 25a for the stable crack tip speed $v_c = 1.47c_S$ which is the lowest stable intersonic crack tip speed attained in the experiments of Rosakis et al. (1999). Daphalapurkar et al. (2007) numerically calculated the lowest sustained crack tip speed to be $1.66c_S$ using GIMP method and the corresponding shear stress pattern is shown in Fig. 25b. In both Fig. 25a and Fig. 25b shear Mach waves are clearly visible with the orientation angles $\beta = 43^\circ$ for the experimental results

Fig. 25 Enlarged view of fringes around an intersonic steady state crack **a** experimental shear stress pattern in Rosakis et al. (1999), **b** numerical prediction of shear stress pattern in Daphalapurkar et al. (2007), **c** PD results of particle velocity distribution



(Rosakis et al. 1999) and $\beta = 37^\circ$ for the numerical results (Daphalapurkar et al. 2007). In our PD simulations, we found the lowest sustained intersonic crack tip speed to be $v_c = 1.57c_S$ for the case of impact speed of 10.2 m/s with an interface toughness of 1.2 Pa.m. In Fig. 25c, the particle velocity contours for the corresponding results are shown using aforementioned parameters. The shear Mach waves with an inclination of $\beta = 40^\circ$ are seen in Fig. 25c. There is a good match of the shear Mach waves obtained in the current study with both the experimental pattern in Rosakis et al. (1999) and the numerical prediction in Daphalapurkar et al. (2007). The inclination of the shear Mach angles in Figs. 25a–c are found to be in agreement with the calculated shear Mach wave angles using Eq. 26 with the calculated crack tip speeds. Based on the comparison of the shear Mach waves in Figs. 25a–c, it can be concluded that the bond based PD with modified PMB constitutive law successfully captures intersonic crack propagation.

6 Conclusions

In this study, shear dominated dynamic crack propagation in the interface of bonded two homogeneous elastic plates is investigated by implementing the bond-based Peridynamic Theory in FEA using ABAQUS. The effect of impact speed and interface toughness on the crack tip speeds and the particle velocity contours is studied. The following conclusions are reached:

1. Both sub-Rayleigh and intersonic crack growth is simulated with bond-based PD with distinguishing particle velocity contours. The Mach waves are clearly identified in the particle velocity magnitude fields of intersonic cracks.
2. For most (G_{II} , V_i) combinations, cracks are observed to start and continue throughout the entire sim-

ulation in the sub-Rayleigh speed regime (at high G_{II} or low V_i) or the intersonic regime (at low G_{II} or high V_i).

3. At a critical impact speed of 10.1 m/s ($G_{II} = 1.2$ Pa.m) or a critical fracture toughness of 2.5 Pa.m (at $V_i = 15$ m/s), the cracks initiate at a sub-Rayleigh speed and transition to an intersonic speed of $1.57c_S$ with a mother-daughter (Burridge–Andrews) crack mechanism. The particle velocity concentration observed at this transition subsequently follows the intersonically growing crack tip at the Rayleigh wave speed, c_R . The gradual rise time of the crack tip speed in the transition regime in our simulations is in good agreement with the experimental results of Rosakis et al. (1999) and Samudrala et al. (2002).
4. The lowest sustained intersonic crack tip speed is found to be $1.57c_S$. Sustained crack tip speeds between $1.57c_S$ and c_D are obtained in the intersonic speed regime for low values of fracture toughness or high values of impact speed. The crack tip speed range between c_R and $1.57c_S$ is found to be unstable.
5. For a shorter pulse duration time of $4 \mu\text{s}$, the crack tip speed jumps to an intersonic speed above $\sqrt{2}c_S$ and then reduces to $\sqrt{2}c_S$ for $5 \mu\text{s}$. However, the crack tip speed is not maintained and the crack tip speed rapidly decelerates to c_R similar to the observations of Needleman (1999).
6. Our observations are in qualitative agreement with the experimental results of Rosakis et al. (1999) and numerical results of Needleman (1999), Abraham and Gao (2000) and Daphalapurkar et al. (2007) and show that bond-based PD can successfully simulate shear-dominated dynamic crack growth.

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